

# First Machine Learning project with Python

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Linear Regression Step-by-Step for Beginners

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### Enlaces:

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# Today's Journey

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1. What is Machine Learning?
2. The Dataset
3. The process to build the model
4. Visualization Results and Predictions
5. Conclusion

# What is Machine Learning?

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- A way for computers to learn from data
- Used to make predictions or decisions
- Examples:
  - Netflix recommendations
  - Spam detection

# The Dataset

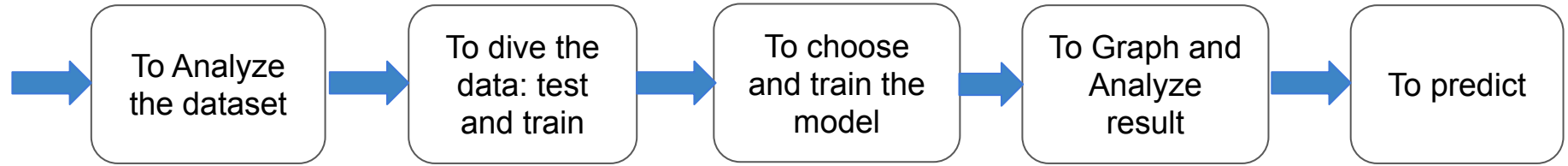
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- In this example, we will use a simple dataset.
- “Hours Studied” vs “Exam Score”

| Hours Studied | Exam Score |
|---------------|------------|
| 1             | 20         |
| 2             | 27         |
| 3             | 34         |
| 4             | 43         |

# Process

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# Train-Test split

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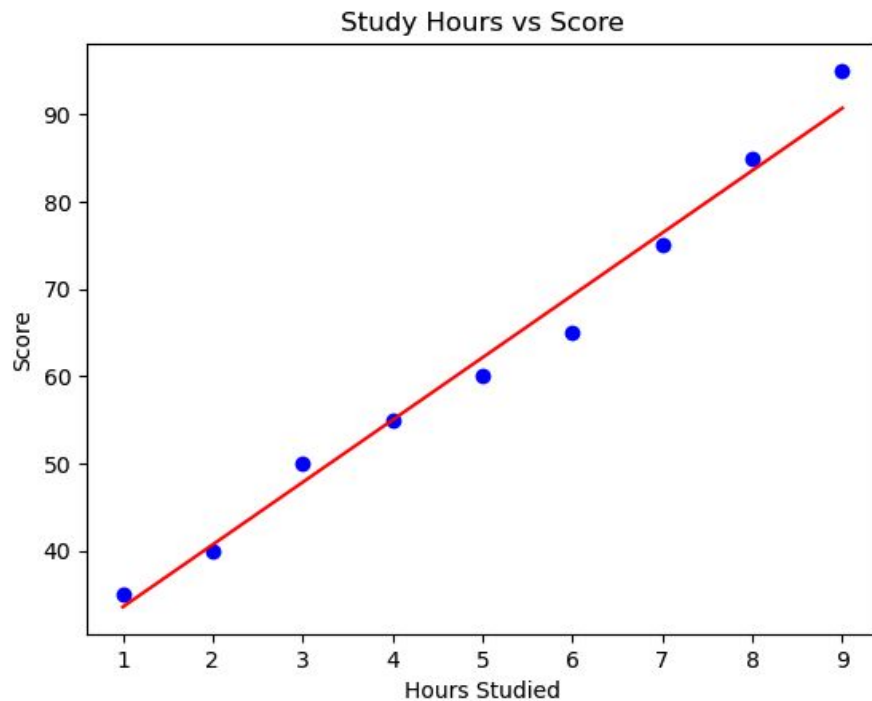
- To divide the dataset:
  - Training set: 80%
  - Test set: 20%
- Is to use to evaluate how well the model work

```
# Step 3: Split into Train-test  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)  
  
X_train
```

# Model Visualization

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- Why is important the visualization?





# Making Predictions

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- Once trained, the model can predict new data
- Example:
  - If a student studies 6 hours ...
  - Model might predict a score of 69

```
# Step 6: Predict new values
hours_pd = pd.DataFrame([[6]], columns=["Hours"])
predicted_score = model.predict(hours_pd)

predicted_score
```

- Github:
  - [https://github.com/GaryBriceno/ml\\_examples/blob/main/beginners/01\\_first\\_project.ipynb](https://github.com/GaryBriceno/ml_examples/blob/main/beginners/01_first_project.ipynb)

# Summary

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- We built a simple Linear Regression Model
- Used Python libraries:
  - Pandas
  - Scikit-Learn
  - Matplotlib
- Visualized and made predictions
- Next:
  - Using a web application and deploy
  - Adding more complexity

# Gracias